JAVASCRIPT

JavaScript Uses

- Respond to user actions (click, input, submit)
- Change HTML and CSS dynamically
- Validate forms
- Create animations
- Fetch data from servers (APIs)
- Build full applications (web, mobile, backend)

Where JavaScript runs

- **Browser**
  - Chrome, Firefox, Edge
  - Used for frontend development
- **Server**
  - Using **Node.js**
  - Used for backend development

Var , let and Const

Feature	var	let	const
Scope	Function	Block	Block
Re-declare	✅ Yes	❌ No	❌ No
Reassign	✅ Yes	✅ Yes	❌ No
Use Today	❌ No	✅ Yes	✅ Yes

Data Types

Primitive Data Types :

1. String
2. Number
3. Boolean
4. Undefined
5. Null
6. Bigint
7. Symbol

Data Types

Non Primitive Data Types :

1. Objects
2. Arrays
3. Functions

Arithmetic Operators

Operator	Name	Example	Result
+	Addition	10 + 5	15
-	Subtraction	10 - 5	5
*	Multiplication	10 * 5	50
/	Division	10 / 5	2
%	Modulus (Remainder)	10 % 3	1
++	Increment	a++	a = a + 1
--	Decrement	a--	a = a - 1

The comparison Operators

Operator	Name	Example	Result
`==`	Equal (value only)	`5 == "5"`	true
`===`	Strict Equal (value + type)	`5 === "5"`	false
`!=`	Not Equal	`5 != 3`	true
`!==`	Strict Not Equal	`5 !== "5"`	true
`>`	Greater than	`10 > 5`	true
`<`	Less than	`5 < 10`	true
`>=`	Greater than or Equal	`10 >= 10`	true
`<=`	Less than or Equal	`5 <= 10`	true

The logical operators

Operator	Name	Description
&&	Logical AND	True if **both** conditions are true
||	Logical OR	True if one of the conditions are true
!	Logical NOT	Reverses the result

The Assignment Operators

Operator	Name	Example	Meaning
=	Assignment	`x = 10`	Assign 10 to x
+=	Add & Assign	`x += 5`	`x = x + 5`
-=	Subtract & Assign	`x -= 3`	`x = x - 3`
*=	Multiply & Assign	`x *= 2`	`x = x * 2`
/=	Divide & Assign	`x /= 4`	`x = x / 4`
%=	Modulus & Assign	`x %= 3`	`x = x % 3`

Functions

A **function** is a block of reusable code that performs a specific task.
It runs **only when it is called**.

Why we use functions

- Avoid code repetition
- Improve readability
- Easy maintenance
- Modular programming

Function Declaration

```
function functionName(parameters) {
  // code
  return value;
}
```

Arrow Functions (ES6)

```
const functionName = (parameters) => {
  return value;
};
```

Parameters

```
function greet(name) {
  console.log("Hello " + name);
}

greet("Naresh");
greet("Student");
```

```
function area(length, width) {
  return length * width;
}

console.log(area(10, 5)); // 50
```

Return Values

```
function sum(a, b) {
  return a + b;
}

let result = sum(5, 6);
console.log(result); // 11
```

Types of Function

Type	Syntax	Hoisting	Usage
Declaration	`function f(){}`	Yes	General use
Expression	`const f = function(){}`	No	Variable-based
Arrow	`const f = () => {}`	No	Modern / React

if, else if, else in JavaScript

```
if (condition) {
   // runs if condition is true
} else if (anotherCondition) {
   // runs if above condition is false and this is true
} else {
   // runs if all conditions are false
}
```

switch Statement in JavaScript

```
switch (expression) {
    case value1:
        // code
        break;
    case value2:
        // code
        break;
    default:
        // code
}
```

For loop

```
for (initialization; condition; increment/decrement) {
    // code
}
```

While loop

```
while (condition) {
   // code
}
```

Do While Loop

```
do {
   // code
} while (condition);
```

For of Loop

```
for (let value of iterable) {
    // code
}
```

For in Loop

```
for (let key in object) {
    // code
}
```

To create project

npm create vite@latest my-app

cd my-app

npm install

npm run dev